Diabetes Prediction Using Machine Learning

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1. Executive Summary

This document reproduces and organizes the work contained in the submitted Jupyter Notebook. It covers library setup, dataset loading and validation, exploratory data analysis, missing-value treatment, feature engineering, train-test splitting, Random Forest model construction and training, performance evaluation, confusion-matrix and ROC analysis, overfitting assessment, feature importance, patient-level predictions, model limitations, the final model conclusion, and an example prediction for a new patient.

The notebook uses a Random Forest Classifier inside a scikit-learn Pipeline. The pipeline performs median imputation followed by Random Forest classification. Three interaction features are engineered before model training.

2. Environment and Libraries

The notebook uses Python 3.12.13 and the following libraries/components:

- NumPy 2.0.2
- Pandas 2.3.3
- Matplotlib 3.10.0
- Seaborn 0.13.2
- Scikit-learn 1.6.1
- Pipeline and SimpleImputer from scikit-learn
- RandomForestClassifier from scikit-learn
- Classification and evaluation metrics from scikit-learn

3. Dataset Loading and Validation

The notebook loads the diabetes CSV dataset from its Kaggle input path. The loaded dataset contains 768 rows and 9 columns.

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age
- Outcome

Outcome is the target variable: 0 = No Diabetes and 1 = Diabetes.

The notebook validates the expected columns and confirms that no unexpected columns are present. It also validates that Outcome contains only 0 and 1.

4. Data Quality Analysis

Standard null-value checking reports zero conventional missing values in all columns.

The notebook then identifies medically implausible zero measurements in Glucose, BloodPressure, SkinThickness, Insulin, and BMI and treats these zeros as missing measurements for modeling.

Zero counts reported by the notebook:

Feature	Zero Count	Zero Percentage
Insulin	374	48.698%
SkinThickness	227	29.557%
BloodPressure	35	4.557%
BMI	11	1.432%
Glucose	5	0.651%

Duplicate-record analysis found 0 duplicate records, so no rows were removed.

5. Statistical Summary

The notebook computes descriptive statistics for all numeric columns and adds a Range column (maximum minus minimum). The main reported ranges are shown below.

Feature	Mean	Range
Pregnancies	3.845	17.000
Glucose	120.895	199.000
BloodPressure	69.105	122.000
SkinThickness	20.536	99.000
Insulin	79.799	846.000
BMI	31.993	67.100
DiabetesPedigreeFunction	0.472	2.342
Age	33.241	60.000
Outcome	0.349	1.000

6. Target Distribution

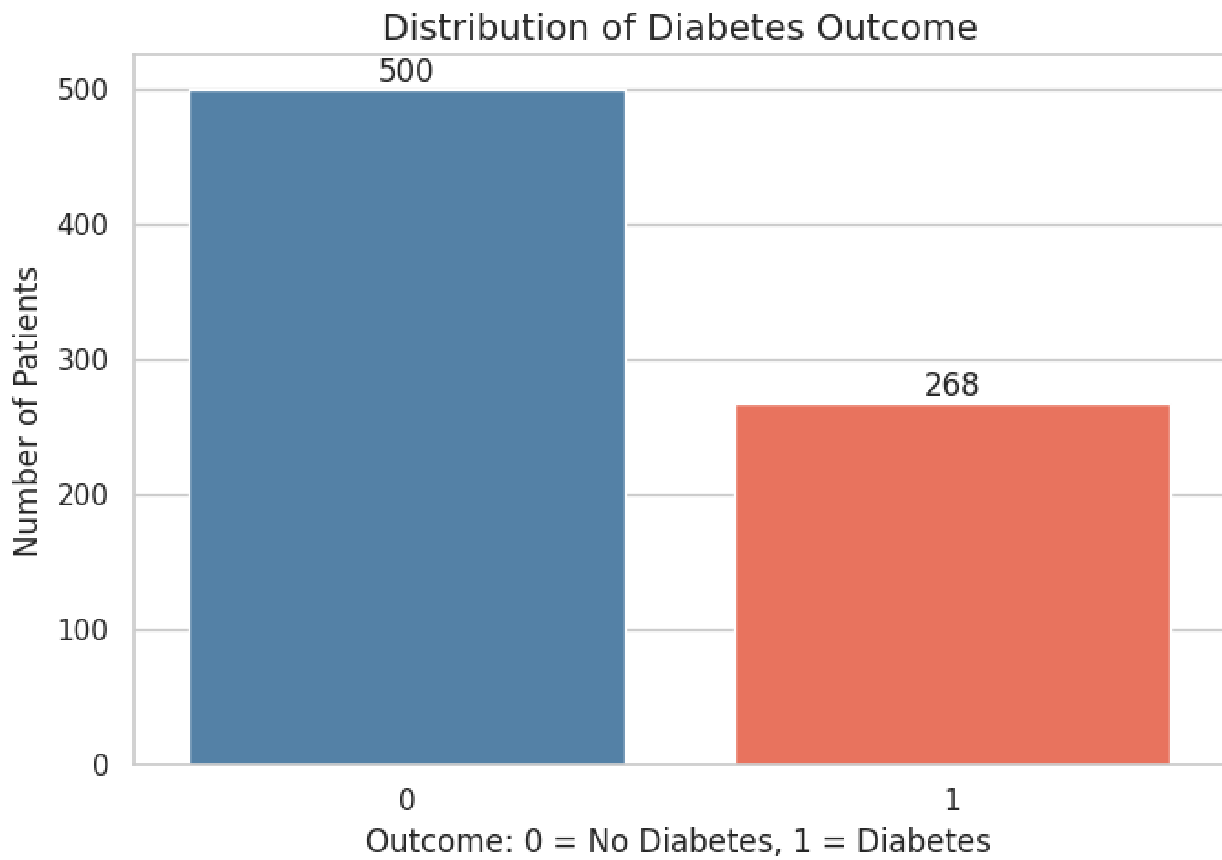
Outcome	Meaning	Count	Percentage
0	No Diabetes	500	65.104%
1	Diabetes	268	34.896%

The notebook describes the dataset as having moderate class imbalance because the minority class represents 34.90% of the data.

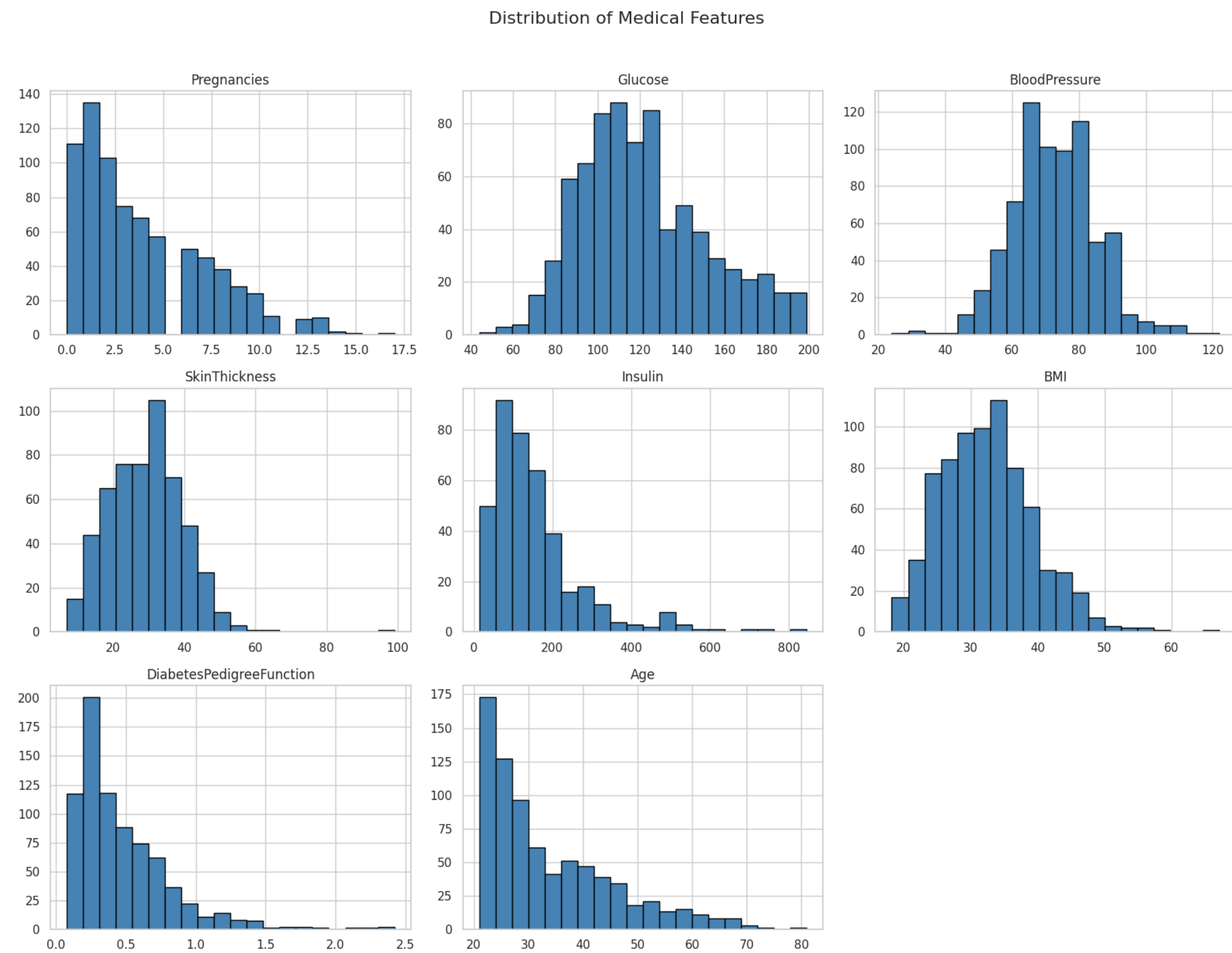
7. Exploratory Data Visualizations

The following figures are inserted from the notebook's embedded PNG outputs without editing the image content.

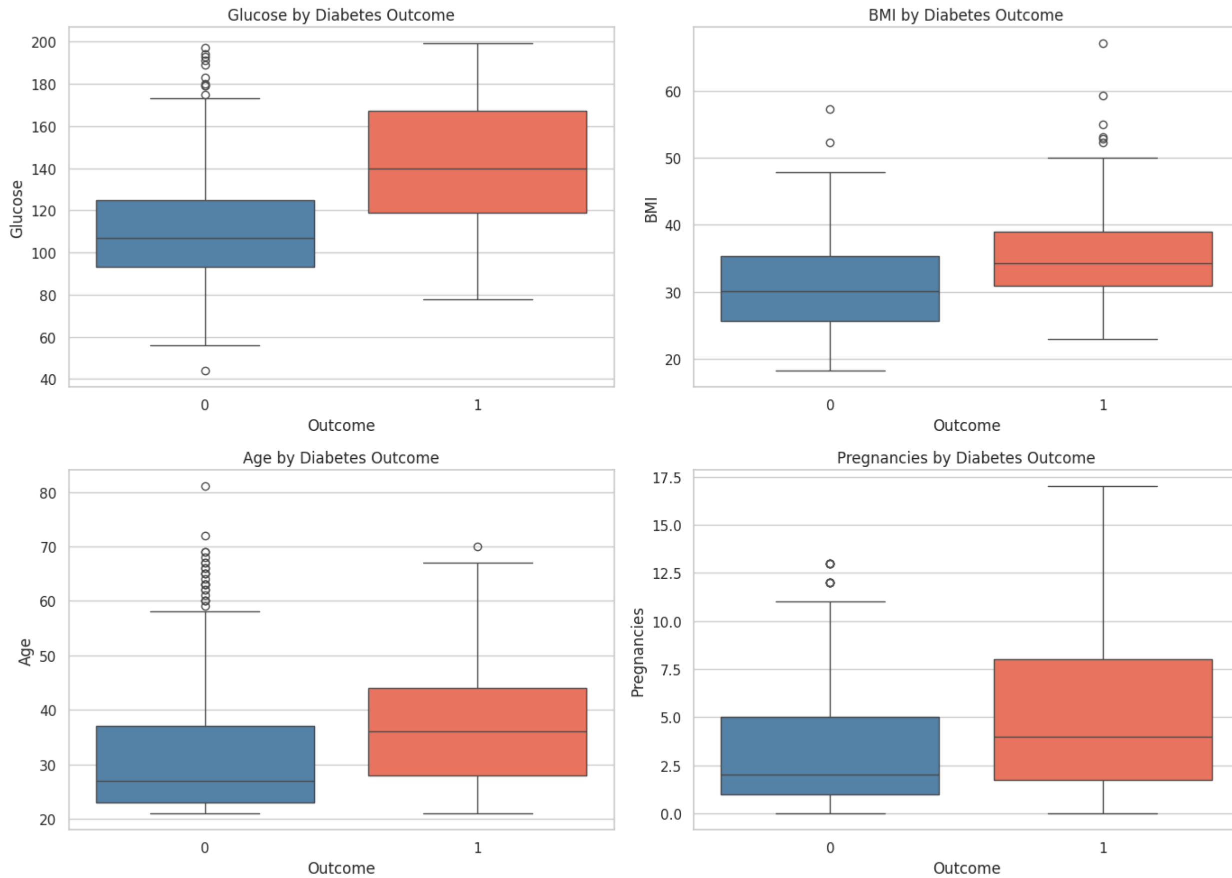
Diabetes outcome distribution



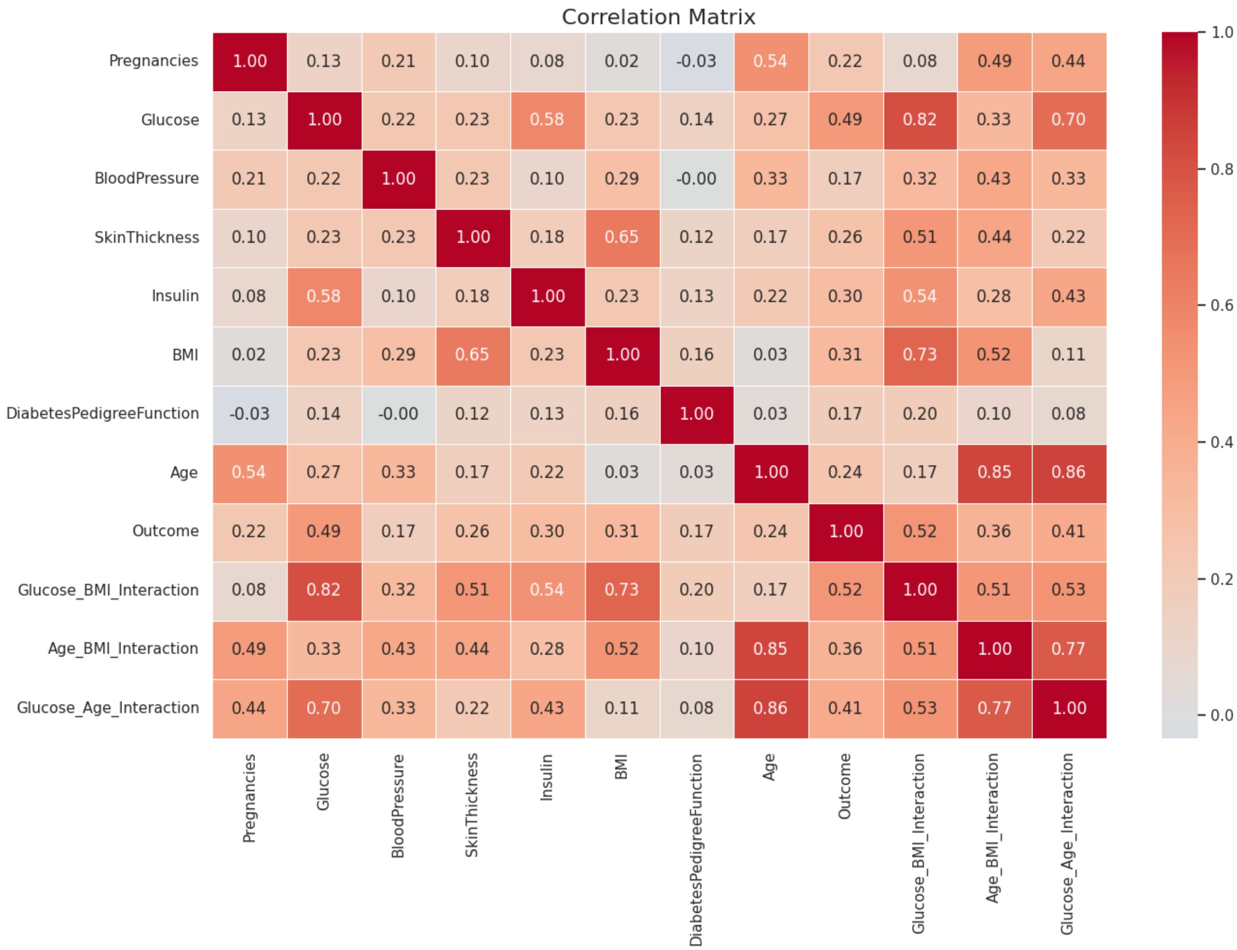
Distribution of medical features



Selected feature distributions by diabetes outcome



Correlation matrix



8. Missing-Value Conversion and Feature Engineering

A separate modeling DataFrame is created. Zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI are replaced with NaN. Median imputation is then performed later inside the modeling Pipeline, using medians calculated from the training data.

Three interaction features are created:

- Glucose_BMI_Interaction = Glucose × BMI
- Age_BMI_Interaction = Age × BMI
- Glucose_Age_Interaction = Glucose × Age

9. Correlation Analysis

The strongest absolute correlation with Outcome is reported as Glucose_BMI_Interaction.

Feature	Correlation with Outcome
Glucose_BMI_Interaction	0.519
Glucose	0.495
Glucose_Age_Interaction	0.409

Age_BMI_Interaction	0.363
BMI	0.314
Insulin	0.303
SkinThickness	0.259
Age	0.238
Pregnancies	0.222
DiabetesPedigreeFunction	0.174
BloodPressure	0.171

10. Train-Test Split

- Total records: 768
- Training records: 576 (75%)
- Testing records: 192 (25%)
- random_state = 42
- stratify = y

The resulting training distribution is 375 No Diabetes and 201 Diabetes. The testing distribution is 125 No Diabetes and 67 Diabetes.

11. Random Forest Model

The selected model is a Random Forest Classifier in a Pipeline.

Parameter	Value
n_estimators	500
max_depth	6
min_samples_split	8
min_samples_leaf	4
max_features	sqrt
class_weight	balanced
random_state	42
n_jobs	-1

The pipeline order is: median imputation → Random Forest classification. The notebook explains that Random Forest was selected for nonlinear relationships, feature interactions, robustness relative to a single decision tree, no need for numerical scaling, and feature-importance output.

12. Model Training and Imputation

The model is fitted only on X_train and y_train. The imputer statistics are calculated from the training data, which the notebook explicitly notes helps prevent information leakage.

Feature	Training Median Used
Pregnancies	3.000
Glucose	117.000
BloodPressure	72.000
SkinThickness	29.000
Insulin	125.000
BMI	32.400

DiabetesPedigreeFunction	0.384
Age	30.000
Glucose_BMI_Interaction	3796.200
Age_BMI_Interaction	1000.500
Glucose_Age_Interaction	3481.500

13. Model Performance

Metric	Training	Testing
Accuracy	0.852	0.750
Precision	0.744	0.620
Recall	0.881	0.731
F1-Score	0.806	0.671
ROC-AUC	0.948	0.826

Testing classification report:

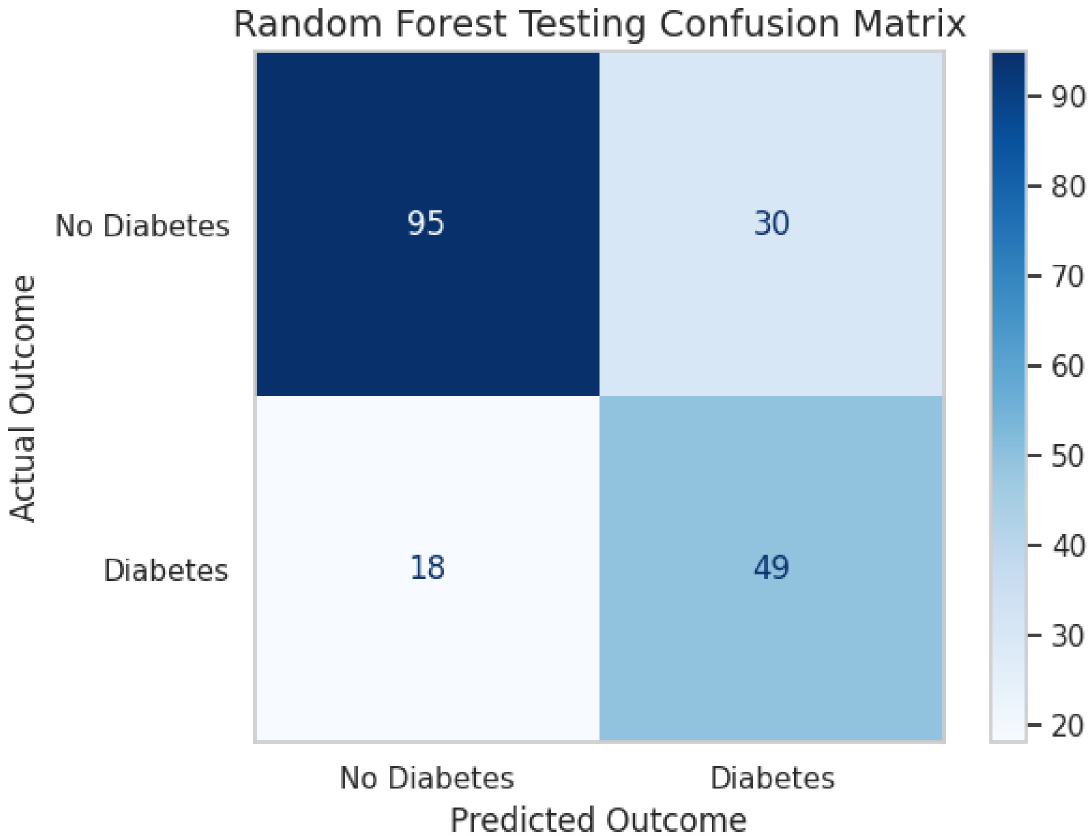
Class	Precision	Recall	F1-score	Support
No Diabetes	0.84	0.76	0.80	125
Diabetes	0.62	0.73	0.67	67
Accuracy		0.75		192
Macro avg	0.73	0.75	0.73	192
Weighted avg	0.76	0.75	0.75	192

14. Performance Comparison Visualization

15. Confusion Matrix

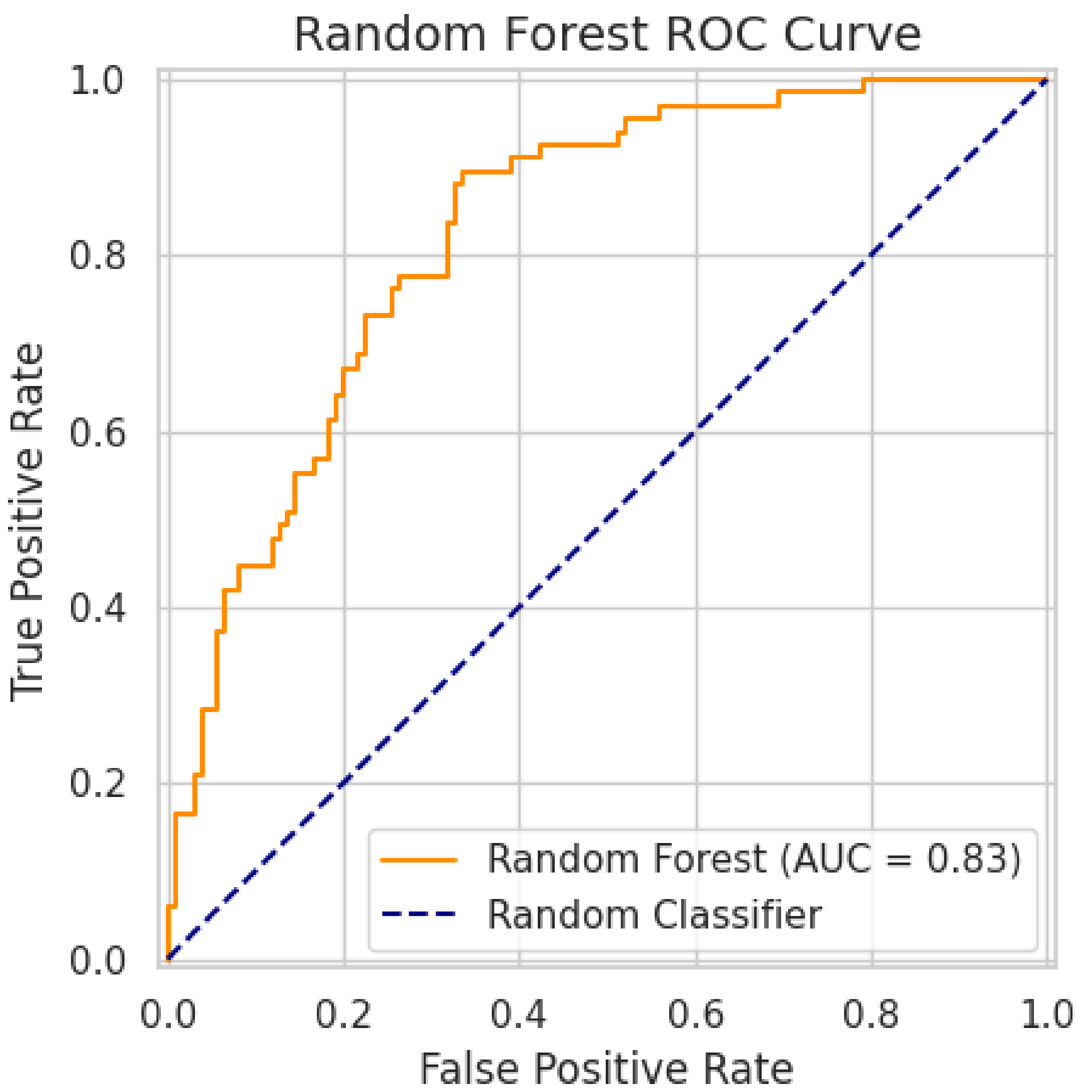
Actual	Predicted No Diabetes	Predicted Diabetes
No Diabetes	95	30
Diabetes	18	49

Therefore: TN = 95, FP = 30, FN = 18, TP = 49.



16. ROC Curve

Testing ROC-AUC reported by the notebook: 0.826.



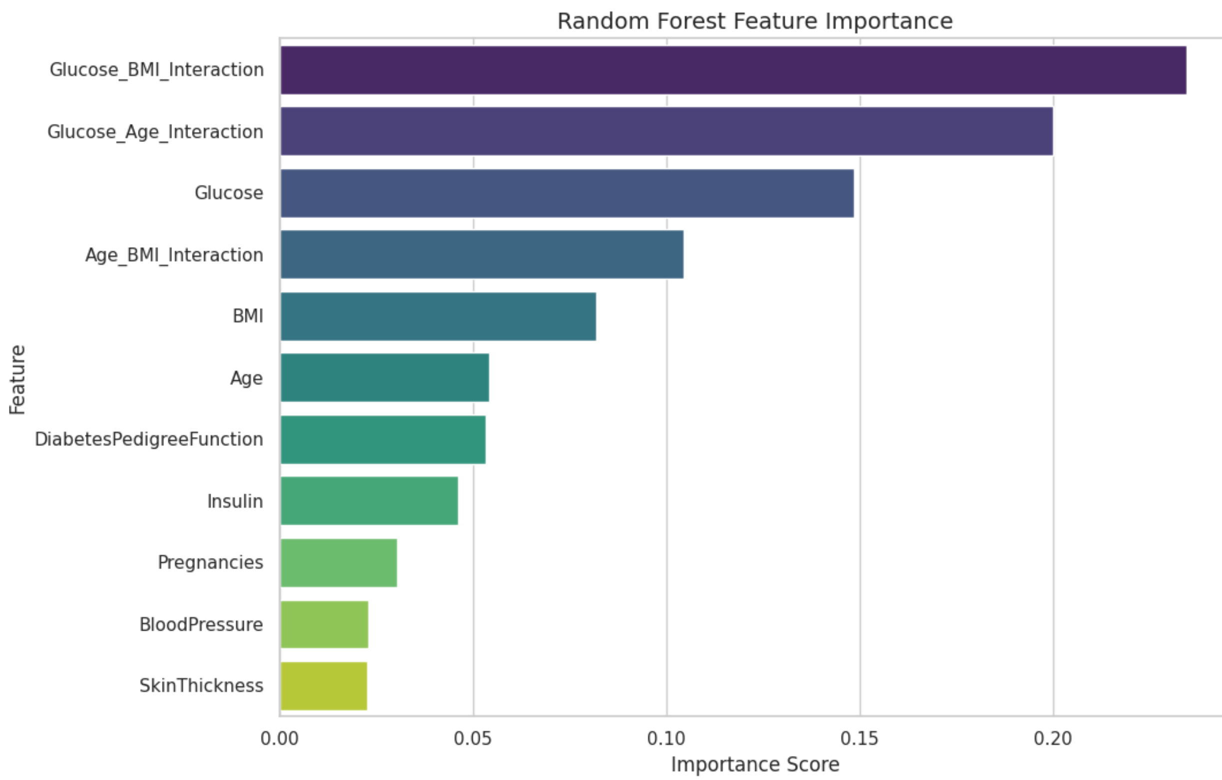
17. Overfitting / Underfitting Analysis

- Training accuracy: 0.852
- Testing accuracy: 0.750
- Accuracy gap: 0.102
- Training F1-score: 0.806
- Testing F1-score: 0.671
- F1 gap: 0.135
- Notebook conclusion: the model shows signs of overfitting because training performance is considerably higher than testing performance.

18. Feature Importance

Rank	Feature	Importance
1	Glucose_BMI_Interaction	0.2346
2	Glucose_Age_Interaction	0.2001
3	Glucose	0.1487
4	Age_BMI_Interaction	0.1047
5	BMI	0.0821

The notebook explicitly cautions that feature importance represents predictive influence and does not establish medical causation.



19. Patient-Level Testing Results

The notebook creates a patient-level testing table containing the original features, actual outcome, predicted outcome, probabilities for both classes, human-readable labels, and a Prediction_Correct flag.

Measure	Value
Total Test Predictions	192
Correct Predictions	144
Incorrect Predictions	48
Correct Prediction Percentage	75.00%
Incorrect Prediction Percentage	25.00%

The notebook reports 48 incorrectly classified testing records and displays the corresponding patient-level rows.

20. Primary Performance Metric

The notebook identifies recall as the most important primary metric for this diabetes prediction problem.

Recall = True Positives / (True Positives + False Negatives). The testing recall is 0.731, with 18 false negatives. The notebook explains why false negatives are especially important in a medical screening context.

21. Model Performance Interpretation

- Testing accuracy: 0.750
- Testing precision: 0.620
- Testing recall: 0.731
- Testing F1-score: 0.671
- Testing ROC-AUC: 0.826
- Recall interpretation: reasonable recall, but some patients with diabetes are still missed.
- F1 interpretation: moderate balance between precision and recall.
- ROC-AUC interpretation: good class-separation ability.
- Fit interpretation: signs of overfitting.

22. Model Limitations

- Evaluation uses a single train-test split, so results may change with another split.
- Zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI were assumed to represent missing values.
- Median imputation may not accurately represent every patient's actual missing measurement.
- Random Forest feature importance indicates predictive influence but not medical causation.
- The dataset may not represent all age groups, locations, ethnicities, or demographic backgrounds.
- The final class prediction uses a 0.50 probability threshold.
- Only the medical features available in the dataset were used.
- The model has not been externally validated on a different patient population.
- The model is intended for educational analysis and not clinical diagnosis.

23. Final Model Conclusion

The notebook concludes that the Random Forest model achieved testing accuracy of 0.750, precision of 0.620, recall of 0.731, F1-score of 0.671, and ROC-AUC of 0.826.

The notebook recommends evaluating suitability primarily using recall, false negatives, F1-score, and the gap between training and testing performance. It also states that cross-validation, threshold optimization, hyperparameter tuning, and external validation would be required before practical application.

24. New Patient Demonstration

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
3	145	78	30	130	32.5	0.450	42

Engineered features for the demonstration patient:

- Glucose_BMI_Interaction = 4712.500
- Age_BMI_Interaction = 1365.000
- Glucose_Age_Interaction = 6090

Notebook prediction:

- Predicted class: 1
- Predicted outcome: Diabetes
- Probability of No Diabetes: 28.31%
- Probability of Diabetes: 71.69%

The notebook clearly labels this prediction as an educational demonstration and not a medical diagnosis.